VAYAS

Age-Stratified Evaluation of Speech Recognition

A portable auditing protocol, demonstrated on Indian languages

Project brief v1.0 • Shubham Gupta, Amity University • Paper 1 of the Vyaskosh series

In one line: This paper establishes how to measure whether speech recognition works for old people — in any language — and shows, using Indian languages, that it largely does not.

1. Objective

Primary. To establish a portable methodology for age-stratified evaluation of automatic speech recognition — specifying how to construct, stratify, and evaluate elderly-speaker benchmarks in *any* language — and to demonstrate it through the first such benchmark and audit of state-of-the-art ASR systems for elderly speakers of Indian languages.

Sub-objectives.

1. Specify the *construction protocol*: recruitment, consent, elicitation design, minimum viable scale, and transcription conventions for age-typical disfluency.
2. Specify the *stratification scheme* across age, language, family, acoustic condition, and speech type.
3. Specify *joint metric decomposition* separating genuine acoustic degradation from morphological metric artifact.
4. Build a *portable error taxonomy* of age-related failure modes.
5. *Instantiate*: two Indian languages across two families, auditing released systems zero-shot.
6. *Release* dataset, evaluation code, protocol document, and replication template.

Positioning. The methodology is the contribution; India is the existence proof. Population ageing is global and no language has age-stratified ASR evaluation — a group in Nigeria, Indonesia or Peru faces the identical problem with none of this data. Indian languages are chosen as a deliberately difficult instantiation: two distinct families, multiple scripts, a steep resource gradient. *Never claim universality as an achieved result; claim portability of the protocol, and demonstrate it.*

2. Core Reading

[v] = citation verified **[?]** = surfaced via AI-assisted search, *verify before citing*

A. Audit method templates — read these first

- **Buolamwini & Gebru (2018)**, FAccT. *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification.* MIT Media Lab. **The structural blueprint** — built a benchmark, trained nothing, audited existing commercial systems, changed industry practice. **[v]**
- **Koenecke, Nam, Lake, . . . Rickford, Jurafsky & Goel (2020)**, PNAS 117(14):7684–7689. *Racial disparities in automated speech recognition.* Stanford. Audited five commercial ASR systems on 42 white / 73 black speakers, 19.8 h — corpus **matched on speaker age and gender**. The control-matching precedent; adopt this design. **[v]**
- Raji & Buolamwini (2019), AIES. *Actionable Auditing.* MIT. Follow-up study on audit impact. **[?]**
- Prinos, Patwari & Power (2024), FAccT. *Speaking of accent: a content analysis of accent misconceptions in ASR research.* Names the unstated, untested generalization assumption. **[v]**

B. Age-stratified ASR — global, non-Indic

- Wilpon & Jacobsen (1996), ICASSP 1:349–352. *A study of speech recognition for children and the elderly.* Foundational. **[v]**
- Vipperla, Renals & Frankel (2008), Interspeech, 2550–2553. *Longitudinal study of ASR performance on ageing voices.* SCOTUS corpus; WER rises *gradually* with speaker age. **[v]**
- Vipperla, Renals & Frankel (2010), EURASIP J. Audio Speech Music Process. *Ageing voices: the effect of changes in voice parameters on ASR performance.* ≈9% absolute WER gap; F0, jitter, shimmer, harmonicity and CPP all differ significantly. Your acoustic mechanism. **[v]**
- **Pellegrini et al. (2012)**, Springer LNCS. *Impact of Age in ASR for the Elderly: Preliminary Experiments in European Portuguese.* Bands 60–90; **41% relative (11.9% absolute) WER increase** from 60–65 to 81–86. **Closest non-English age-banding precedent — model your bands on this.** **[v]**
- Wolters, Georgila, MacPherson & Moore (2009), ACM TACCESS 2(1). *Being old doesn't mean acting old: older users' interaction with spoken dialogue systems.* **[v]**
- **Kulkarni, Kulkarni, Trancoso & Couceiro (2024)**, Interspeech. *Unveiling Biases while Embracing Sustainability.* Reportedly found **better** ASR performance for older speakers on read English speech. **This is the counterexample to your hypothesis — read it before recording anyone.** **[?]**
- Hu et al. (2024), arXiv:2407.13782. Self-supervised models and features for dysarthric and elderly speech. **[?]**
- LT-EDI-2025 Tamil elderly / vulnerable-speaker shared task. Closest Indic precedent — chase this one hard. **[?]**

C. Indic context and benchmark construction

- **Joshi et al. (2025)**, Interspeech. *SRUTI* (rural Bhojpuri women). **Your scale template**: 444 utterances, 51 speakers, ≈72 transcribed minutes. **[v]**
- Javed et al. (2024). *IndicVoices.* Speaker demographics, elicitation design, 22 languages. **[v]**
- Bhogale et al. (2023), Interspeech. *Vistaar*, arXiv:2305.15386. **[v]**
- Dhasmana, Srivastava & Chiang (2026), VarDial, arXiv:2601.04373. *Dialect Matters.* Differentiate against. **[v]**
- Juvekar et al. (2026), arXiv:2605.13087. *Vividh-ASR.* Complexity tiers; a 244M model matched 769M under R-MFT. **[v]**
- Basu et al. (2026), arXiv:2606.09345. Region-specific Indic ASR; fine-tuning used as a controlled probe. **[v]**

D. Metrics and release practice

- **Thennal D K, James, Gopinath & Ashraf K (2025)**, Findings of NAACL, 4941–4950. *Advocating Character Error Rate for Multilingual ASR Evaluation.* **Grounds the WER–CER decomposition. Non-negotiable citation.** **[v]**
- Gebru et al. *Datasheets for Datasets.* **[?]**
- Mitchell et al. (2019), FAccT. *Model Cards for Model Reporting.* **[?]**
- Littell et al. (2017), EACL, 8–14. *URIEL and lang2vec.* Typological distance, if predictors are analysed. **[v]**

E. Systems under audit — train nothing

- Whisper (Radford et al., 2022); MMS (Pratap et al., 2023); USM (Zhang et al., 2023); Meta Omnilingual ASR (2025). **[v]**
- IndicWhisper / IndicWav2Vec / IndicConformer (AI4Bharat); Sarvam Saarika. **[v]**

3. Metrics to Report

Metric	Role
WER	Comparability with all prior ASR work
CER	Primary for the Dravidian language; morphology-robust
WER − CER divergence	**Separates real acoustic degradation from metric artifact**
Task success rate	Did the utterance resolve to the correct welfare scheme
Entity accuracy	Scheme names, numbers, dates, place names
Sub / Ins / Del ratios	Error-type diagnosis
Speaking rate × WER	Acoustic mechanism of degradation
Per-speaker distribution	**Worst case and variance, not corpus aggregate**

Stratify every metric across: age band — **control (<60)**, 60–69, 70–79, 80+; language and language family; speech type (read vs. spontaneous); acoustic condition (quiet vs. ambient noise); and gender.

Statistical reporting. Confidence intervals and effect sizes across groups — never bare point estimates; bias audits are held to this standard. *Aggregate WER hides the single speaker for whom the system fails completely, and that speaker is the point of the paper.*

4. Method

1. **Protocol first, recording second.** Write the construction protocol *before* recording anyone, as though a stranger abroad will execute it. Justify every choice in language-general terms — a prompt chosen because it works in Hindi breaks the moment someone runs it in Yoruba. This discipline is what makes the portability claim true rather than decorative.
2. **Recruitment and consent.** Elderly (60+) participants, rural where possible, with a documented ethics and consent procedure — itself part of the contribution, since social barriers to recording elderly rural speakers are the main reason this data does not already exist.
3. **Matched control cohort.** A non-elderly cohort recorded under *identical* prompts and conditions. Without it you can report an absolute number but cannot claim a gap. Follow Koenecke et al.'s matching design. **Non-negotiable.**
4. **Elicitation.** Read speech (controlled, comparable) *and* spontaneous speech (realistic), in the welfare and service domain: pensions, healthcare, agriculture, documentation.
5. **Scale target.** ≈50 speakers and ≈1 transcribed hour per language, across two languages spanning Indo-Aryan and Dravidian. Calibrated against SRUTI — achievable in a semester, defensible in review.
6. **Transcription conventions.** Explicit handling of age-typical disfluency, hesitation and self-repair; inter-annotator agreement reported.
7. **Zero-shot audit.** Run all systems exactly as released. Train nothing. The contribution is measurement.
8. **Error taxonomy.** Categories defined to be portable across languages, not Hindi-specific.
9. **Release.** Dataset, evaluation code, protocol document, replication template. The artifact is the point.

Open gates before committing. (1) Confirm no portable age-stratified ASR auditing protocol already exists in any language. (2) Confirm real access to ≈50 elderly speakers per language. *Everything above is hypothetical until there are recordings.*